OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T5:
RetBot, distance control

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1 Preparations

1.1. (aV) P controller in the time domain

(aV) What is the equation that describes a digital P controller in the time domain? [1]

$$x_a[t] = K_P \cdot e[t] \tag{1.1}$$

1. $x_a(t)$: Input variable of the P controller
2. K_p: Proportional coefficient
3. $y(t)$: Control output of the P controller

1.2. (bV) Transfer function of open loop P controller

(bV) What is the transfer function of the open loop with P controller?
Which characteristic shows $G_o(s)$? [1]

The transfer function $G_o(s)$ is obtained using the block diagram determined in the fourth experiment as follows:

$$G_o(s) = K_P \cdot \frac{-K_I}{T_1 \cdot s^2 + s} \tag{1.2}$$

$G_o(s)$ exhibits IT1 behavior.

1.3. (cV) Transfer function of distance control with P controller

(cV) What is the transfer function of the distance control with P controller?
Which characteristic shows $G_W(s)$? [1]

By inserting 1.2, we obtain the following for $G_W(s)$:

$$G_W(s) = \frac{G_o(s)}{1 + G_o(s)} = \frac{1}{1 + \frac{-T_1}{K_P \cdot K_I} \cdot s^2 - \frac{1}{K_P \cdot K_I} \cdot s} \tag{1.3}$$

The closed-loop system $G_W(s)$ exhibits PT_2 behavior.

1.4. (dV) Reasonable damping ratio D for the closed control loop

(dV) Select a reasonable damping ratio D for the closed control loop and calculate the controller parameter K_P so that the desired/selected damping ratio is achieved. [1]

The damping ratio is set to D = 1 (aperiodic limit case) to ensure a fast, overshoot-free settling. This allows the robot to reach the target distance as quickly as possible without having to move backward. The standard mathematical form of a PT2 system is:

$$G_s = \frac{K_P}{\frac{1}{\omega_0^2} \cdot s^2 + \frac{2D}{\omega_0 \cdot s} + 1} \tag{1.4}$$

Rearranging 1.4 yields:

$$\frac{1}{\omega_0^2} = -\frac{T_1}{K_P \cdot K_I} \rightarrow \omega_0 = \sqrt{-\frac{K_P \cdot K_I}{T_1}} \tag{1.5}$$

$$\frac{2D}{\omega_0} = -\frac{1}{K_P \cdot K_I} \rightarrow \omega_0 = -K_P \cdot K_I \cdot 2 \cdot D \tag{1.6}$$

If we now set Equation 1.6 equal to 1.5, we can solve for K_P.

$$\sqrt{-\frac{K_P \cdot K_I}{T_1}} = -K_P \cdot K_I \cdot 2 \cdot D \rightarrow K_P = \frac{1}{4 \cdot D^2 \cdot T_1 \cdot K_I} \tag{1.7}$$

Using the previously determined parameters (mean value) K_I and T_1,
Left motor:

$$K_{I,L} = 0{,}437\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,L} = 57{,}1\,\mathrm{ms} = 0{,}0571\,\mathrm{s} \tag{1.8}$$

Right motor:

$$K_{I,R} = 0{,}446\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,R} = 52{,}2\,\mathrm{ms} = 0{,}0522\,\mathrm{s} \tag{1.9}$$

the following values are obtained:
Left motor:

$$K_{P,L} = \frac{1}{4 \cdot 1^2 \cdot 0.0571 \cdot 0.437} = 9{,}99\,\%\,\mathrm{cm^{-1}} \approx 10\,\%\,\mathrm{cm^{-1}} \tag{1.10}$$

Right motor:

$$K_{P,R} = \frac{1}{4 \cdot 1^2 \cdot 0.0522 \cdot 0.446} = 10{,}77\,\%\,\mathrm{cm^{-1}} \approx 11\,\%\,\mathrm{cm^{-1}} \tag{1.11}$$

minus werte müssen rauskommen

2 Experiment T5: RetBot, distance control

name sections with title of the task

2.1. e)

(e) Simulate the idealized behavior of the closed control loop in Matlab/Simulink using the controller parameter K_P calculated in the preparation by using the setup from previous experiment. [1]

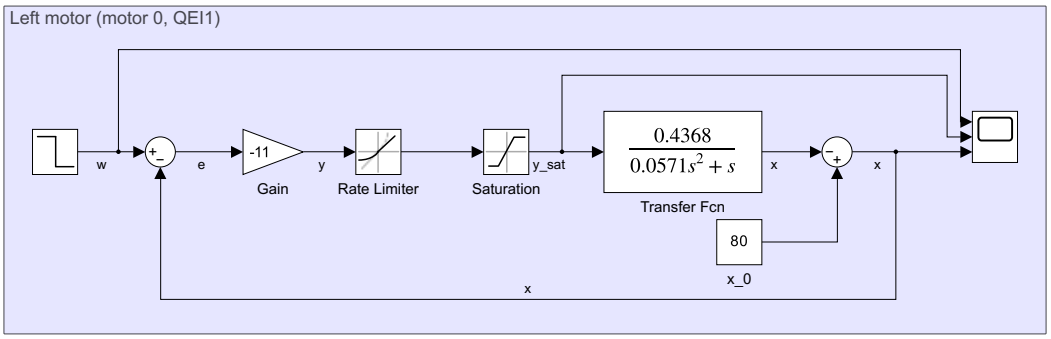


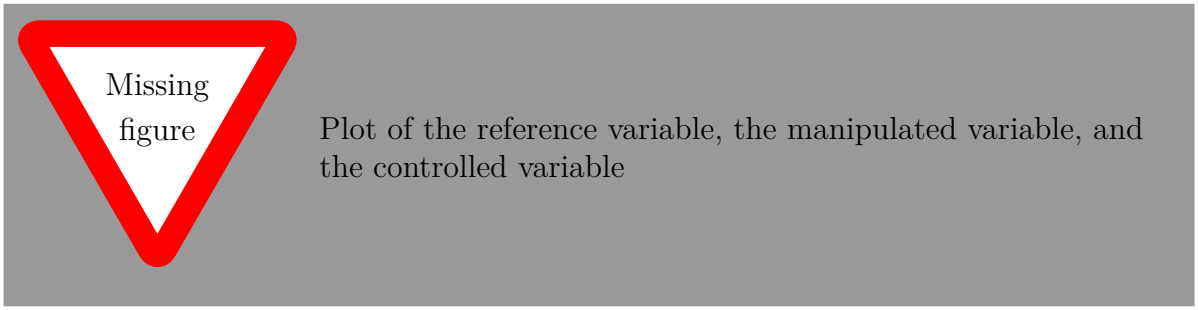
Abbildung 2.1.: Image of the simulink controller

cut of white space from simulink image print on pdf

Figure 2.1 shows the Simulink model used for the simulation of the closed control loop. The model is based on the one from the previous experiment, with the addition of a P controller block. The parameters for the simulation are set according to the values calculated in section 1.4.

2.2. f)

(f) Plot the reference variable, the manipulated variable, and the controlled variable via Matlab (clean graphs, no screenshots from the scope). Is the damping level you selected correct? Interpret the signal curves. [1]



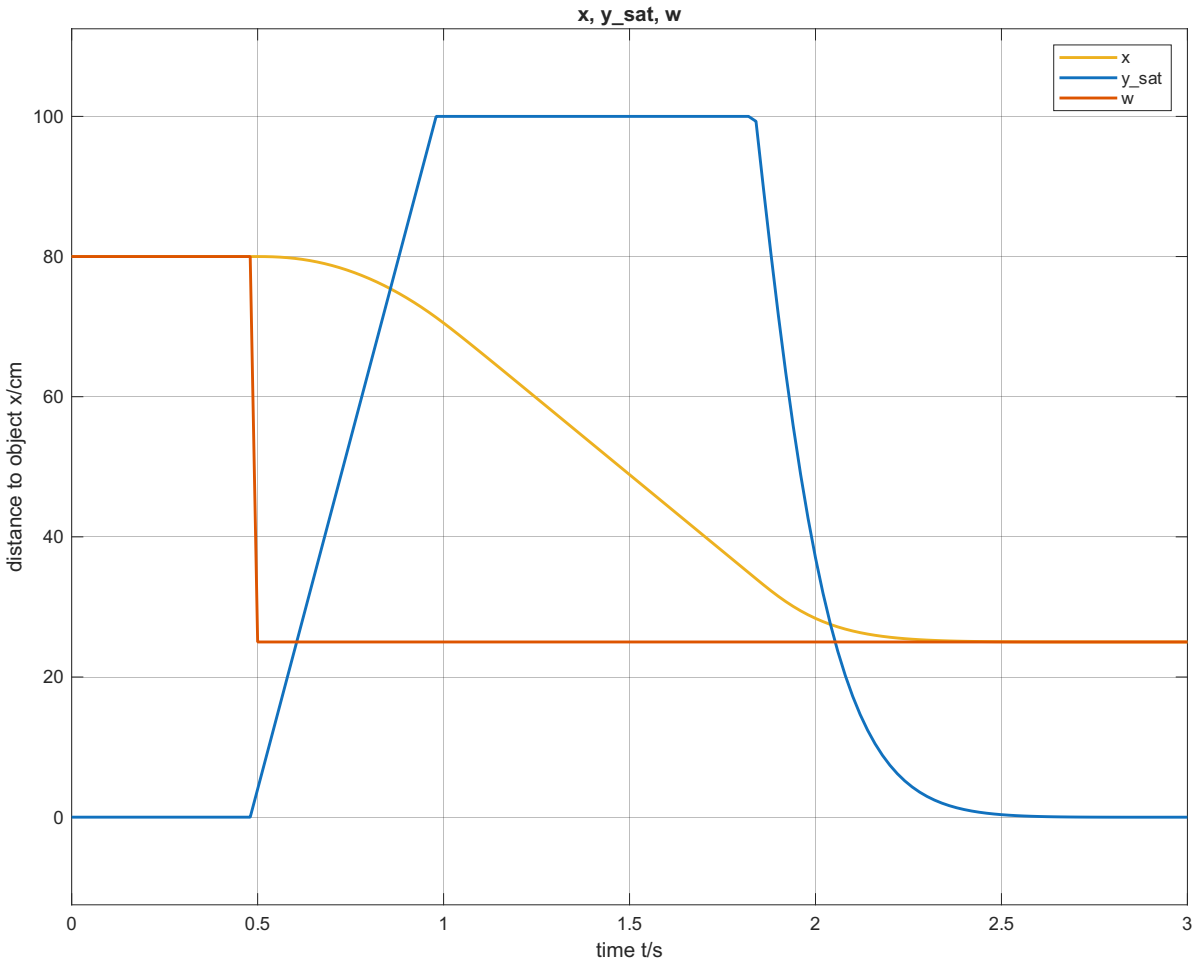


Abbildung 2.2.: Plot of the reference variable, the manipulated variable, and the controlled variable

interpret dampening level

2.3. g)

(g) Implement the P controller for the LEGO robot - see 3.7 for the existing software-based control loop, that can be parameterized by using following commands using Serial.ICE:

- *'1': set K_P*

- *'2': set K_I*
- *'3': set K_D*
- *'4': set desired distance/cm*
- *'5': set K_0*
- *'6': set K_1*

Show the correct implementation by Serial.ICE commands and received data. [1]

explain what has been send

Listing 2.1: Serial.ICE commands send

```
r       -> QEI reset
m0      ->
1-11    -> set K_I to "-11"
20      -> set K_P to "0"
30      -> set K_D to "0"
425     -> set desired distance to "25" cm
50.99   -> set K_0/Motor0 to "0.99" rotations for each rotation
61      -> set K_1/Motor1 to "1" rotation for each rotation
```

The received data from the robot after sending the commands is shown in Listing 2.2.

Listing 2.2: Serial.ICE commands received

```
QEI-Reset done
Motor-state: 0
PID-Kp: -11.000000
PID-Ki: 0.000000
PID-Kd: 0.000000
Desired-dist/cm: 25.000000
K0 set to 0.990000
K1 set to 1.000000
fsample=500Hz
Time/ms;QEI1/steps;QEI2/steps;d/cm;Mot0/%
   0;    0;    0; 81.0;  3
   2;    0;    0; 81.0;  3
   4;    0;    0; 81.0;  3
   6;    0;    0; 81.0;  3
   8;    0;    0; 81.0;  3
  10;    0;    0; 81.0;  3
```

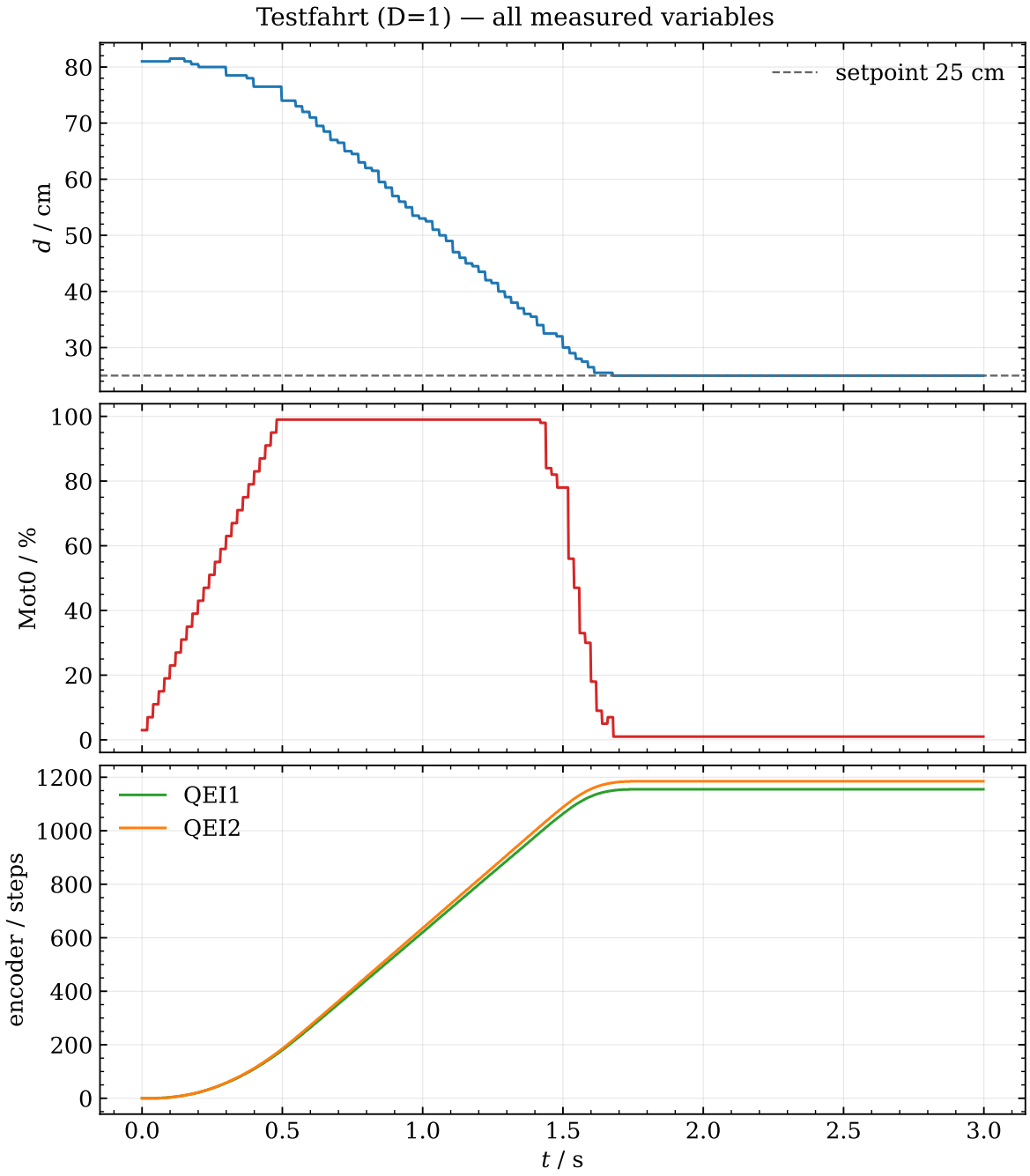
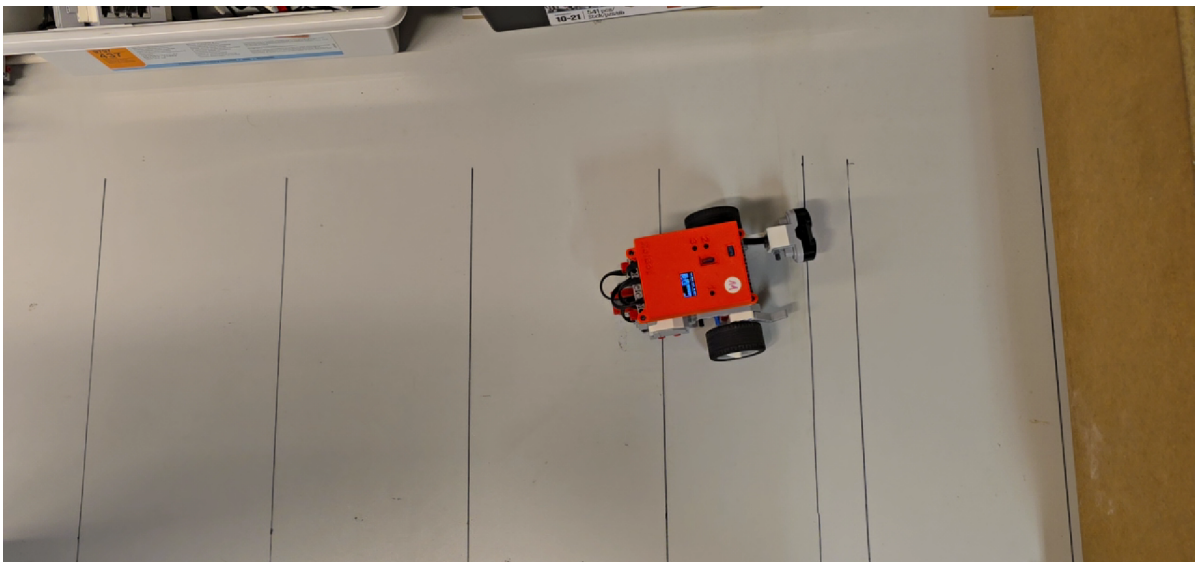


Abbildung 2.3.: QEI und Distance Retrun from Serial.ICE

2.4. h)

(h) Upload a video showing the robot moving towards an obstacle. [1]



https://youtu.be/gG7SXy00EJQ

2.5. i)

(i) Read the recorded data of your test-drive using Serial.ICE and compare it to the simulated results - distance, travelled distance and duty-cycle of motor 0! This has to be done in one graph after importing the data according to the previous experiment! [1]

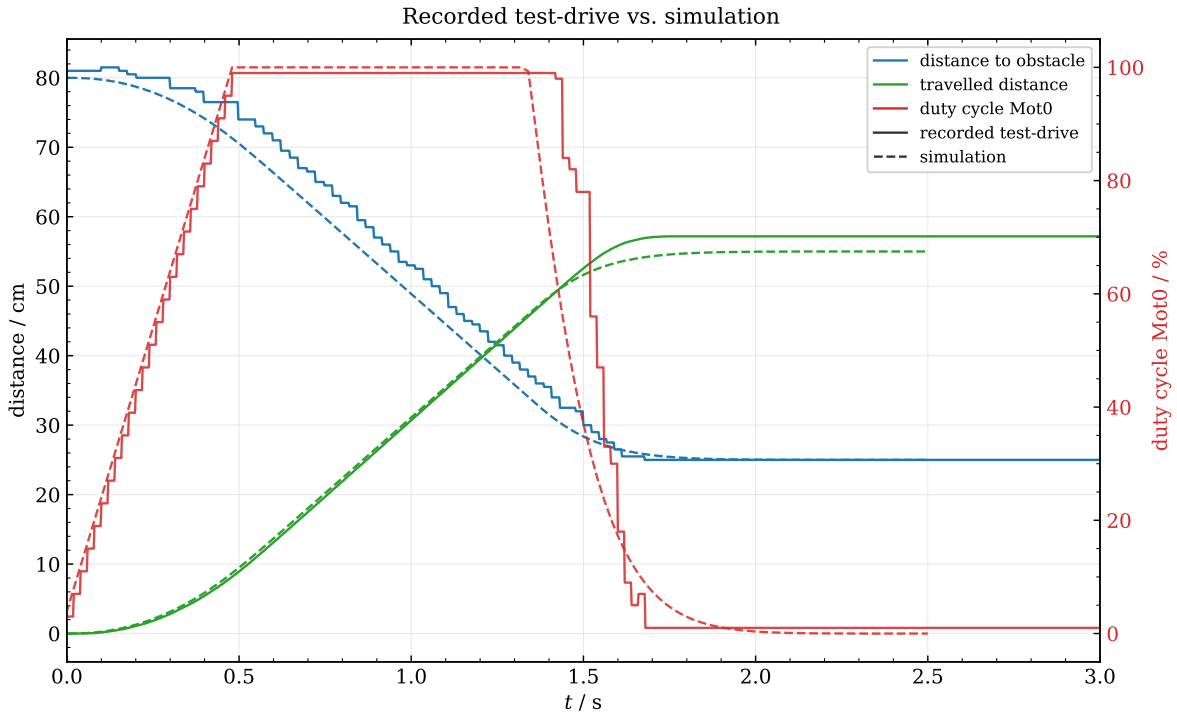


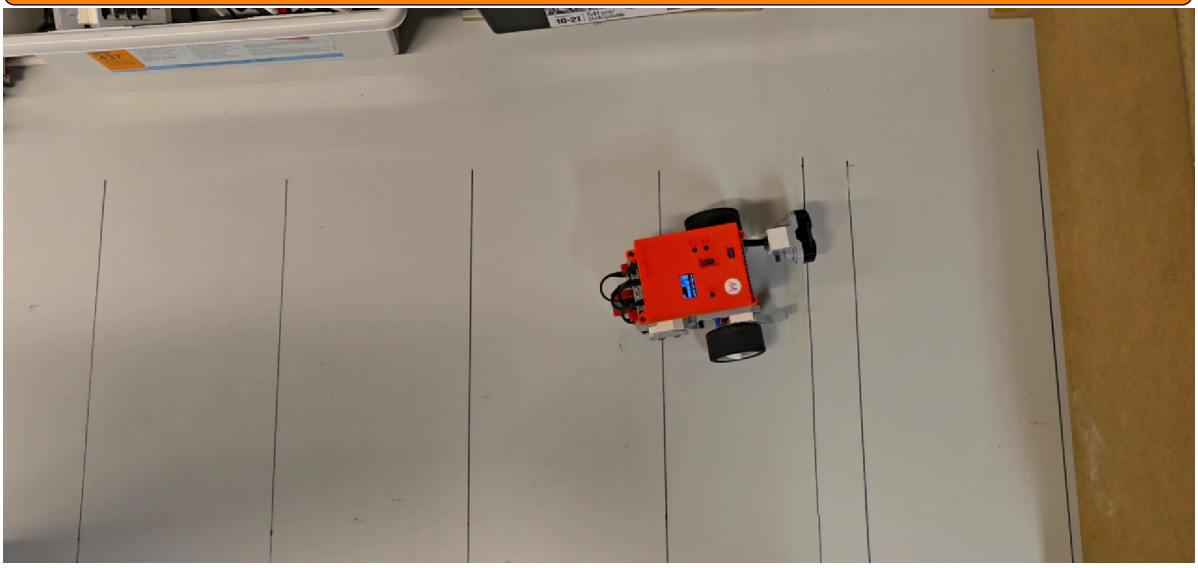
Abbildung 2.4.: Recorded test-drive vs. simulation: distance, travelled distance and duty cycle of motor 0.

compare test to simulation data + error analysis

2.6. j)

(j) Increase K_P, check the influence of the rate-limit-factor in a way that the robot shows periodic oscillations - write down this critical value(s) and upload a video. [1]

explain critical value(s)



https://youtu.be/cy5t8bU4MxE

$$K_{P,R} = \frac{-1}{4 \cdot 0.50^2 \cdot 0.0522 \cdot 0.446} = -42.95 \tag{2.1}$$

With $D = 0.5$ to achieve periodic oscillations, the critical value for K_P is $-42{,}95$
The video shows the robot oscillating around the target distance for this the following Serial.ICE commands were sent:

Listing 2.3: Serial.ICE commands and received datalog

```
r
m0
1-42
20
30
425
50.99
61
```

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

Regensburg, den 2026-05-01,

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Ort, Datum und Unterschrift

Abbildungsverzeichnis

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B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations

Editorial note — Source code omitted

The full source code for the analysis (a Jupyter notebook of this time only about 1000 lines and a simulink model) is ***not*** included in this appendix. Including it as a verbatim listing still caused major build instability again again and was therefore deliberately omitted.
The complete source can be obtained by either of the following means:

- By email request to `janis.graf@st.oth-regensburg.de`
- Via requesting access to the GitHub repository: `https://github.com/janisgra/RTP-T5-SoSe2026.git`

Literatur

[1] Torsten Reitmeier. *Internship in Control Engineering (RTP)*. Praktikumsversuche. 2025.